

SESSION 2

Basic Physics and Electronics in Radiology

Capacitance and Capacitors in X-ray Equipment

Energy storage, charge-discharge behaviour and radiographic applications

Course	RAD-201 Radiation Physics
Date	Oct, 2026
Lecturer	A. A. Zubair
Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. describe the structure and purpose of a capacitor in simple physical terms;
2. define capacitance and explain the meaning of $C = Q/V$;
3. predict how plate area, plate separation and dielectric material affect capacitance;
4. calculate stored electrical energy and explain why high voltage greatly increases capacitor hazard;
5. explain capacitor charging and discharging using the RC time constant;
6. distinguish capacitor voltage from charging or discharging current; and
7. connect capacitor behaviour to X-ray generator smoothing, mobile capacitor-discharge systems, timing/control circuits and equipment safety.

The radiography idea to remember

A capacitor is a device that can **store separated electric charge and electrical energy, then release that energy when the circuit requires it**. In radiographic equipment, this basic behaviour can support high-voltage power supplies, smooth electrical waveforms, provide short bursts of energy, and assist timing or control circuits.

1 Session Orientation: think “store, hold, release”

Session 1 followed electrons through the X-ray tube. This session asks what happens when a circuit needs to **store electrical energy temporarily** before using it. Four questions organize the topic:

1. **How is charge stored?** Opposite charges are separated on two conductors.
2. **How much can be stored?** That ability is described by capacitance.
3. **How fast does the voltage change?** Resistance and capacitance together set the RC time constant.
4. **Why does radiography care?** Capacitors help shape generator output and can retain dangerous energy after the mains supply is removed.



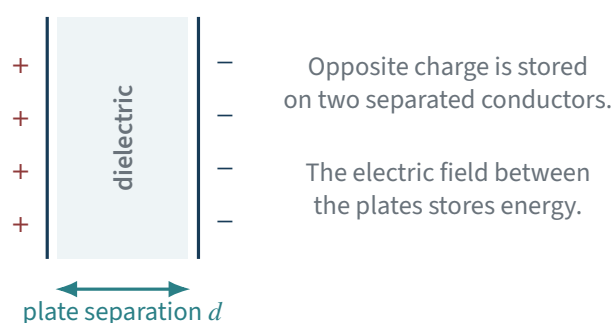
Radiographer's viewpoint

You do not need to service high-voltage capacitors. You do need to understand why generator type changes X-ray output, why a mobile unit can deliver a short high-power exposure from stored energy, and why “switched off” does not automatically mean “electrically safe.”

2 What is a capacitor?

A **capacitor** consists of two conducting surfaces separated by an insulating material called a **dielectric**. When a voltage source is connected, electrons are moved from one plate toward the other through the external circuit. One plate therefore gains excess electrons and becomes negatively charged, while the other plate is left with an equal positive charge.

The dielectric prevents the two plates from making direct electrical contact. Instead of a continuous current crossing the dielectric, an **electric field** exists between the separated charges. Energy is stored in that electric field.



A capacitor is not a battery

Both can store energy, but they do so differently. A battery mainly stores energy chemically and can maintain a source voltage for relatively long periods. A capacitor stores energy in an electric field and can usually charge or discharge much more rapidly.

3 Capacitance: how much charge can be stored?

Capacitance describes how much separated charge a capacitor stores for each volt of potential difference:

$$C = \frac{Q}{V}.$$

Here, C is capacitance in farads (F), Q is charge in coulombs (C), and V is potential difference in volts (V).

A larger capacitance means that, at the same voltage, the device stores more charge. For example, a $100\ \mu\text{F}$ capacitor stores twice as much charge at a given voltage as a $50\ \mu\text{F}$ capacitor.

Charge stored at a given voltage

A $20\ \mu\text{F}$ capacitor is charged to 5 kV. The stored charge is

$$Q = CV = (20 \times 10^{-6})(5000) = 0.10\ \text{C}.$$

The calculation does not mean that charge crossed the dielectric. It means that equal and opposite charges have accumulated on the two conductors.

3.1 What controls capacitance?

For a simple parallel-plate capacitor,

$$C = \frac{\epsilon_0 \epsilon_r A}{d}.$$

The equation is useful because it describes the physical design of the device.

Design factor	If it increases	Simple explanation
Plate area, A	Capacitance increases	More conductor surface is available to hold separated charge.
Plate separation, d	Capacitance decreases	A larger gap weakens the interaction between the separated charges.
Relative permittivity, ϵ_r	Capacitance increases	A suitable dielectric reduces the effective field opposition and allows more charge to be stored at the same applied voltage.

Remember the direction

Bigger plates, closer plates and a higher-permittivity dielectric increase capacitance. The dielectric also provides electrical insulation, which is especially important in high-voltage imaging equipment.

4 Energy stored in a capacitor

A charged capacitor stores electrical potential energy:

$$E = \frac{1}{2}CV^2.$$

The most important feature for radiographic equipment is the V^2 term. Stored energy rises very rapidly as voltage increases.

Why high voltage changes the safety picture

Consider a 1 μF capacitor charged to 100 kV:

$$E = \frac{1}{2}(1 \times 10^{-6})(100000)^2 = 5000 \text{ J.}$$

Five kilojoules is substantial stored energy. This illustrates why high-voltage capacitors must be treated as hazardous even after the main power switch has been opened.

Voltage is squared

If capacitance stays unchanged and voltage doubles,

$$E_{\text{new}} = \frac{1}{2}C(2V)^2 = 4E_{\text{old}}.$$

Doubling voltage therefore quadruples stored energy.

Equipment safety

Capacitors may remain charged after the equipment is switched off. Radiographers should never open high-voltage generator compartments or attempt to discharge components. Servicing requires trained personnel, approved isolation procedures and verified discharge of stored energy.

5 Charging: why voltage rises but current falls

Connect an initially uncharged capacitor to a DC source through a resistor. At the instant of connection, the capacitor voltage is approximately zero, so the full supply voltage drives current through the resistor. Charge begins to accumulate on the plates.

As charge accumulates, capacitor voltage rises. That rising voltage increasingly opposes the supply voltage, so the charging current becomes smaller. Eventually the capacitor voltage approaches the supply voltage and the charging current approaches zero.

For a simple RC circuit,

$$V_C(t) = V_0 \left(1 - e^{-t/RC}\right).$$

The important idea is the shape, not memorizing the exponential: **fast change at first, then progressively slower change as the final value is approached.**

6 The RC time constant

The speed of charging and discharging is governed by

$$\tau = RC,$$

where R is resistance and C is capacitance. The unit of RC is seconds.

Elapsed time	Charging: fraction of final voltage reached	Discharging: fraction of initial voltage remaining
1τ	about 63.2%	about 36.8%
2τ	about 86.5%	about 13.5%
3τ	about 95.0%	about 5.0%
5τ	about 99.3%	about 0.7%

Time constant

A $100\ \mu\text{F}$ capacitor charges through a $10\ \text{k}\Omega$ resistor:

$$\tau = RC = (10\,000)(100 \times 10^{-6}) = 1\ \text{s}.$$

After about 5τ , or 5 s, charging is practically complete for most purposes.

What changes the speed?

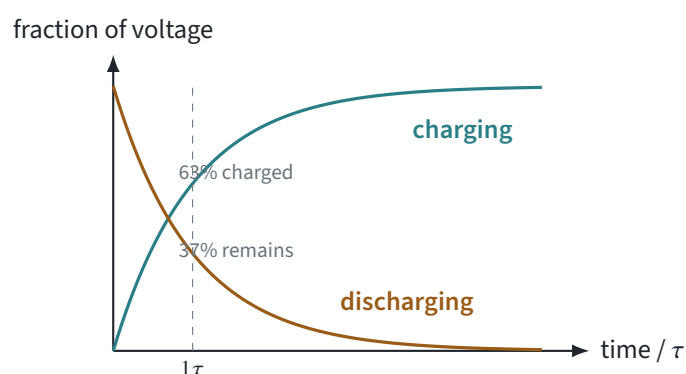
A larger resistance or larger capacitance gives a larger τ , so charging and discharging take longer. A smaller R or C gives a shorter time constant and faster electrical response.

7 Discharging: the stored field gives energy back

When the supply is removed and the capacitor is connected across a load or resistor, stored charge begins to recombine through the external circuit. Capacitor voltage falls according to

$$V_C(t) = V_0 e^{-t/RC}.$$

At the start of discharge, capacitor voltage and discharge current are largest. As energy leaves the capacitor, voltage falls and the current also decreases.



Current and capacitor voltage do not behave the same way

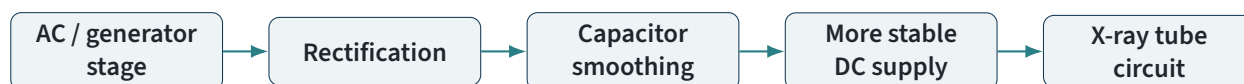
During charging, capacitor voltage rises while charging current falls. During discharging, both capacitor voltage and discharge current decrease in magnitude. Do not memorize only a voltage curve and assume the current follows the same shape and direction.

8 Where capacitors appear in X-ray equipment

The same ability to store and release energy can be used in several different ways depending on where the capacitor is placed in a circuit.

8.1 1. Smoothing generator output

Rectification makes current flow in the desired direction, but the rectified voltage may still fluctuate. A capacitor can charge when the supply voltage is high and release energy when the supply voltage falls. This reduces voltage ripple and helps produce a more stable DC supply.



A more stable tube voltage produces a more consistent X-ray spectrum and output than a strongly fluctuating voltage of the same nominal peak value. Modern generator designs use more complex power electronics, but capacitors remain important energy-storage and filtering components within those systems.

8.2 2. Capacitor-discharge mobile X-ray systems

A capacitor bank can be charged before the exposure and then discharged through the X-ray tube. This allows the system to deliver a short burst of relatively high power even when the available mains supply is limited.

A key consequence is **voltage droop**: as the capacitor gives up charge, its voltage falls. Without compensation, the tube potential can therefore decrease during the exposure. That changes electron energy and can change the X-ray spectrum and output as the exposure proceeds.

Why mobile generators may behave differently

Generator design affects the waveform actually applied to the tube. Two exposures with the same displayed kVp and mAs may not be physically identical if the generators have very different voltage ripple or discharge characteristics. This is one reason radiographers should understand generator type rather than treating console numbers as isolated values.

8.3 3. Timing, integration and control circuits

An RC circuit changes voltage in a predictable time-dependent way. Capacitors can therefore be used in timing, filtering and control circuits. In exposure-control systems, a signal related to detector response can be accumulated or integrated until a threshold is reached and the exposure is terminated. The exact implementation differs among systems, but the core idea is that controlled charge accumulation can be used to make an electrical decision.

8.4 4. Fast switching and pulsed imaging

Capacitors can provide or absorb energy rapidly, which makes them useful in high-speed power and switching circuits. In pulsed fluoroscopy and angiographic equipment, fast electronic control is required to start and stop X-ray production accurately. The capacitor supports the power/control circuitry; the grid or switching devices determine when electron flow is permitted.

9 Capacitors in series and parallel: a useful equipment idea

Large energy-storage systems often use several capacitors rather than one component. Two simple rules help explain why.

Parallel connection

Capacitances add: $C_{eq} = C_1 + C_2 + \dots$. The bank can store more charge at a given voltage.

Series connection

The reciprocal capacitances add: $1/C_{eq} = 1/C_1 + 1/C_2 + \dots$. The equivalent capacitance is lower, but the applied voltage can be shared across multiple components when the design includes appropriate balancing.

Do not reduce this to “series bad, parallel good”

The arrangement depends on the design goal. Parallel connections are useful when greater capacitance is needed; series arrangements can be useful when the circuit must withstand a larger total voltage. High-voltage engineering requires proper voltage sharing and safety design.

10 A radiography framework for capacitor questions

When you meet a capacitor question in imaging physics, ask four things:

1. **What is being stored?** Separated charge and electrical energy.
2. **What sets the amount?** Capacitance and applied voltage.
3. **How quickly does it change?** The circuit resistance and capacitance, summarized by $\tau = RC$.
4. **What is the radiographic purpose?** Energy storage, smoothing, timing, control or rapid power delivery.

Capacitors do not “increase kVp” by themselves

A capacitor stores and releases energy. Whether it smooths voltage, supports a mobile discharge exposure, or forms part of a timing/control circuit depends on its position and function in the generator design.

Key Takeaways

- A capacitor consists of two conductors separated by a dielectric and stores energy in the electric field between separated charges.
- Capacitance is charge stored per unit voltage: $C = Q/V$.
- Larger plate area, smaller plate separation and higher dielectric permittivity increase capacitance.
- Stored energy is $E = \frac{1}{2}CV^2$; high voltage is especially important because the voltage term is squared.
- Charging and discharging are exponential processes governed by the RC time constant, $\tau = RC$.
- Capacitors can smooth generator voltage, store energy for mobile exposures, and support timing, filtering and fast control circuits.
- A charged high-voltage capacitor may remain dangerous after the equipment is switched off.

Key Terms

Capacitor

Device that stores separated charge and electrical energy in an electric field.

Capacitance

Charge stored per unit potential difference, measured in farads.

Dielectric

Insulating material separating capacitor conductors.

RC time constant

$\tau = RC$; measure of how quickly a simple capacitor circuit charges or discharges.

Voltage ripple

Residual variation in a rectified or DC voltage waveform.

Voltage droop

Fall in capacitor voltage as stored charge is released during discharge.

Capacitor-discharge unit

X-ray system that stores electrical energy before releasing it to the tube circuit during exposure.

Self-Assessment Questions

1. Describe the physical structure of a capacitor and state what the dielectric does.
2. What does the equation $C = Q/V$ mean in words?
3. Explain how plate area, plate separation and dielectric permittivity affect capacitance.
4. Calculate the charge stored by a $10\ \mu\text{F}$ capacitor at $2\ \text{kV}$.
5. Why does high voltage make stored capacitor energy especially hazardous?
6. A $47\ \mu\text{F}$ capacitor is connected through $2.2\ \text{k}\Omega$. Calculate τ and estimate the time for near-complete charging.
7. Explain in words why charging current decreases as capacitor voltage rises.
8. Compare capacitor voltage after 1τ during charging and discharging.
9. Explain how a capacitor smooths a rectified generator voltage.
10. What is voltage droop in a capacitor-discharge X-ray unit, and why can it matter to X-ray output?

Series navigation

Session 1 established charge, voltage, current and the electron pathway through the X-ray tube. **Session 2** adds temporary electrical-energy storage and RC behaviour. The next electrical topics can now build on how generator circuits transform, rectify, smooth and control the power delivered to the tube.

